GENERAL ASSEMBLY

GENERALIZED LINEAR MODELS

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Data Science Immersive

LINEAR AND LOGISTIC REGRESSION

- So far, we've learned how to fit linear and logistic regression models.

- Linear and logistic regressions are both types of **linear models**.
  - What do we mean when we say **linear model**?
    - We want to model some outcome by a series of predictors $X_1, \ldots, X_p$, where a one-unit change in an independent variable has a fixed effect on the outcome.
    - You might also hear the phrase **linear in the coefficient**.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 \log(X_1)$$

Z W

LINEAR AND LOGISTIC REGRESSION

- If I want to predict some continuous Y value between $(-\infty, \infty)$, I'll probably use a linear regression model.
- If I want to predict some continuous Y value between (0,1), I'll probably use a logistic regression model.
  - Remember that logistic regression models predict probabilities between 0 and 1 and that we can use these probabilities to classify observations into one of two classes.

threshold = 50%
↑ most common

- What are examples of things we might like to predict that don't fall in either a $(-\infty, \infty)$ or (0,1) range?

GOING BEYOND LINEAR AND LOGISTIC REGRESSION

- What are examples of things we might like to predict that don't fall in either a $(-\infty, \infty)$ or (0,1) range?

- vote for D/R/I $\{0,1,2\}$
- SAT/ACT scores $\{0,1,2,\ldots,36\}$
- stock price $[0,\infty)$
- where a wildfire will occur
- how long until my light bulb burns out $[0,\infty)$

GENERALIZED LINEAR MODELS

GLMs

- This is where **generalized linear models** comes into play.
  - We can create models that take the benefits of being linear in the coefficient and "bend" it to match the range of the *Y* we want to model.
  - In order to make this happen, we need three things:
    - A linear piece.
    - A "bending" piece.
    - A random piece.

GENERALIZED LINEAR MODELS

- Generalized linear models have the following form:

$$Y = g(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p) + \varepsilon$$

bending / link

linear

random component

GENERALIZED LINEAR MODELS

- Generalized linear models have the following form:

$$Y = g\left(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p\right) + \varepsilon$$

- $\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$ is our **linear component**.
  - We want a one-unit change in X to change Y in a certain way.
- $g(\cdot)$ is our **link component**.
  - We need to "bend" our linear component to match the range of Y.
- ε is our **random component**.
  - We need our errors to also match our Y.

GENERALIZED LINEAR MODELS

$$Y = g(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p) + \varepsilon$$

- **Linear regression:**
  - Linear Component:

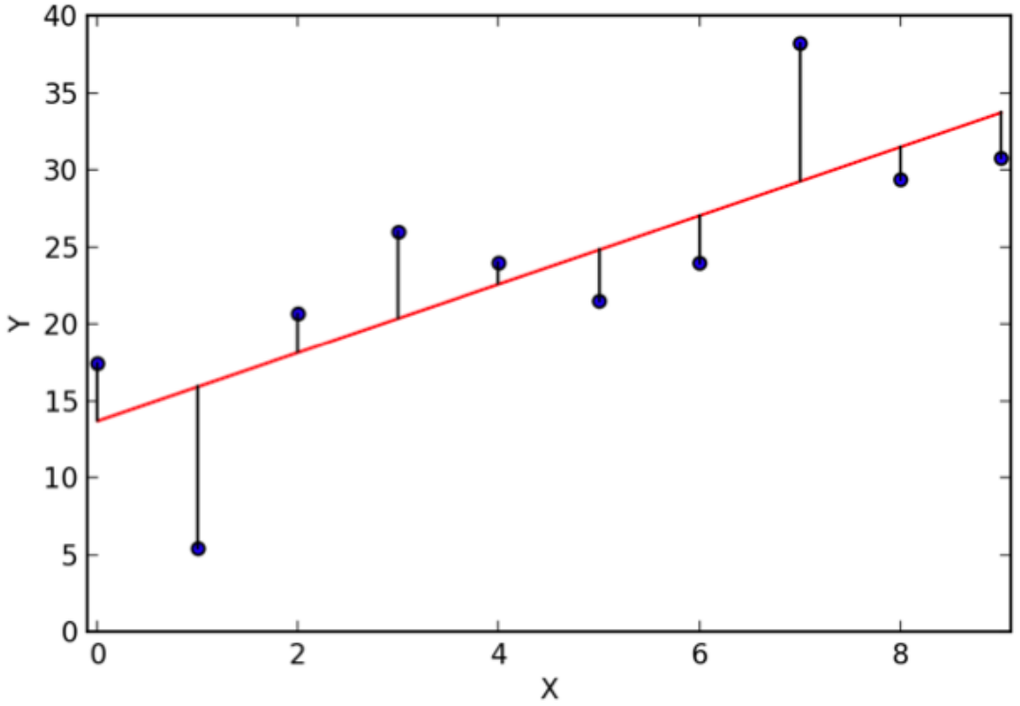
$\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$

  - Link Component:

Identity function $f(x) = x$

  - Random Component:

$\varepsilon \overset{ind}{\sim} N(0, \sigma)$



LINEI
✓ ✓ ✓ ✓ ✓

$\varepsilon_i = y_i - \hat{y}_i$

GENERALIZED LINEAR MODELS

$$Y = g(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p) + \varepsilon$$

- **Logistic regression:**
  - Linear Component:

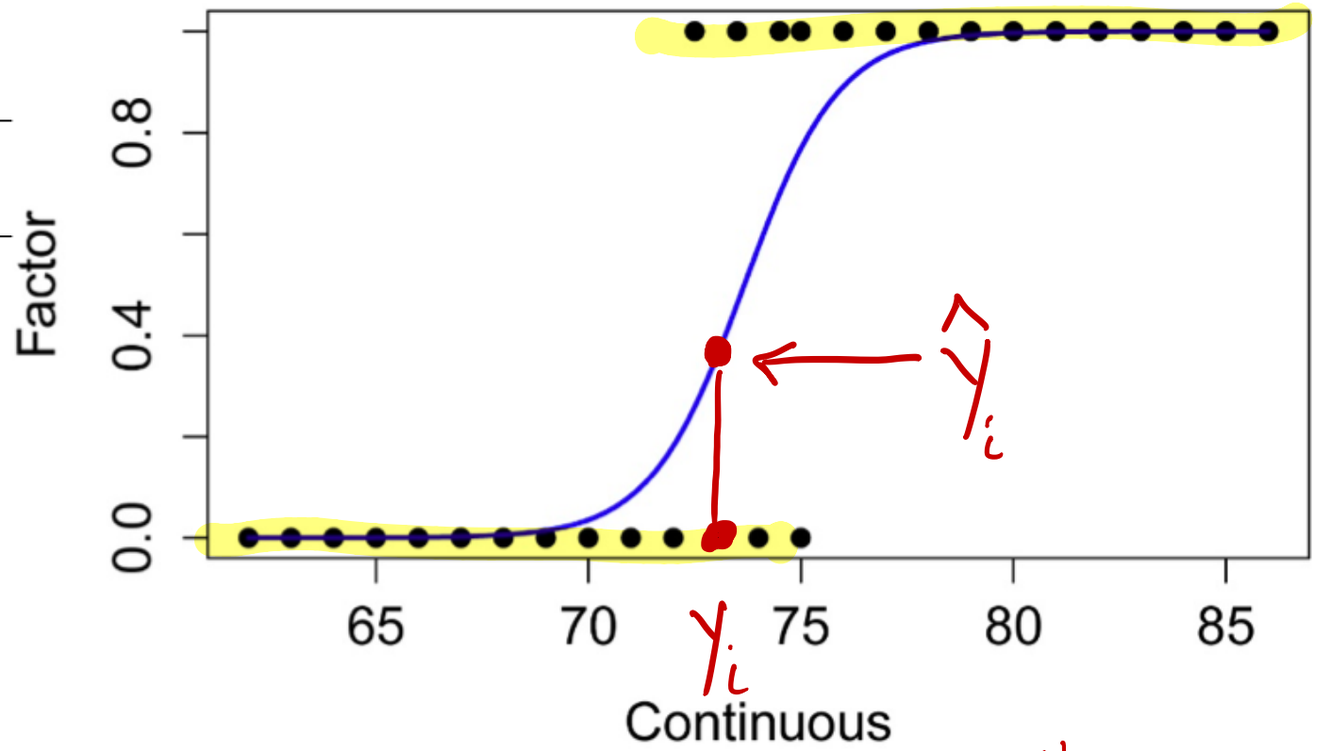
$$\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

  - Link Component:

$$\text{logit}(\cdot)$$

  - Random Component:

$$\varepsilon_i \sim \text{Bernoulli}(p_i)$$



"in"

Predicted: $\hat{y} \in (0, 1)$

Observed: $y \in \{0, 1\}$

GENERALIZED LINEAR MODELS

- Generalized linear models have the following form:

$$Y = g\left(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p\right) + \varepsilon$$

- What does the link function do?
  - It transforms the linear component to match the range of Y.
  - It also affects how we interpret our coefficients.

- What does the random component do?
  - It relates our predicted values $\hat{y}$ to our observed values y.

GENERALIZED LINEAR MODELS

Count

non-negative integer values

- If your Y values are $\{0,1,2,\ldots\}$:
  - Poisson Regression (if $Var[X] \approx E[X]$)
  - Negative Binomial Regression (if $Var[X] \gg E[X]$)

mean ≒ variance

overdispersion

How many units do I sell?

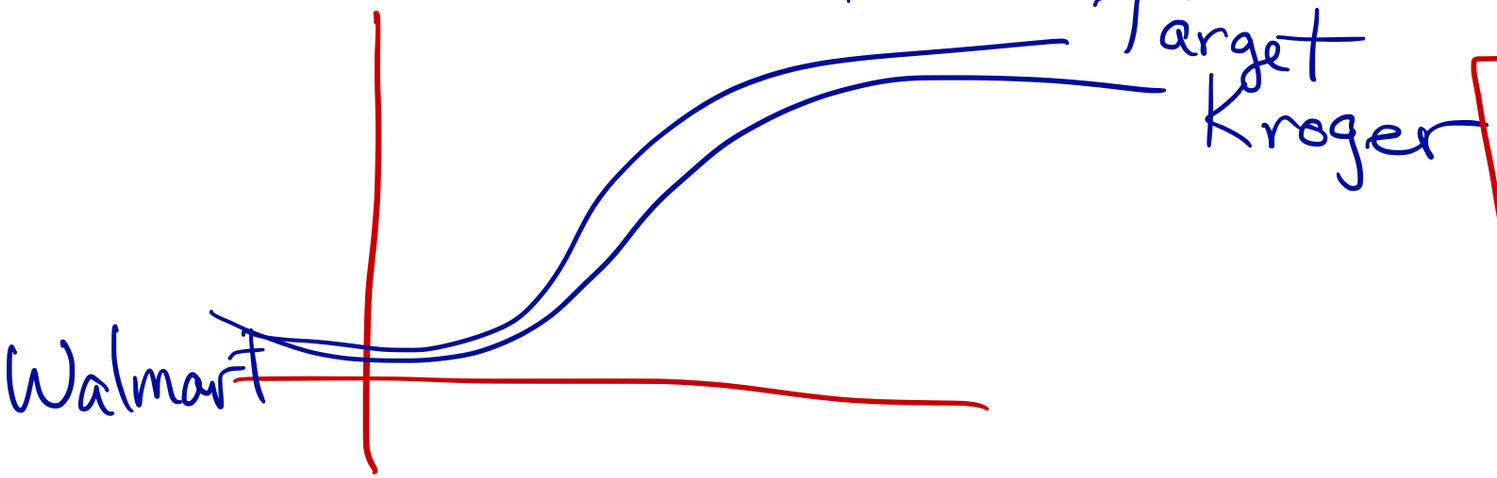
How many customers do I expect to see?

How many buses go by before my bus gets here?

GENERALIZED LINEAR MODELS

- If your Y values are $\{0,1,2 \ldots, k\}$: → fixed number of categories
  - Multinomial Logistic Regression (if categories are unordered)
  - Ordinal Logistic Regression (if categories are ordered)

Will people buy their Jif at Walmart, Kroger, Target?



You watched House of Cards on Netflix. How many stars do you rate it?

GENERALIZED LINEAR MODELS

- If your Y values are $[0, \infty)$: → continuous, non-negative
  - Gamma Regression

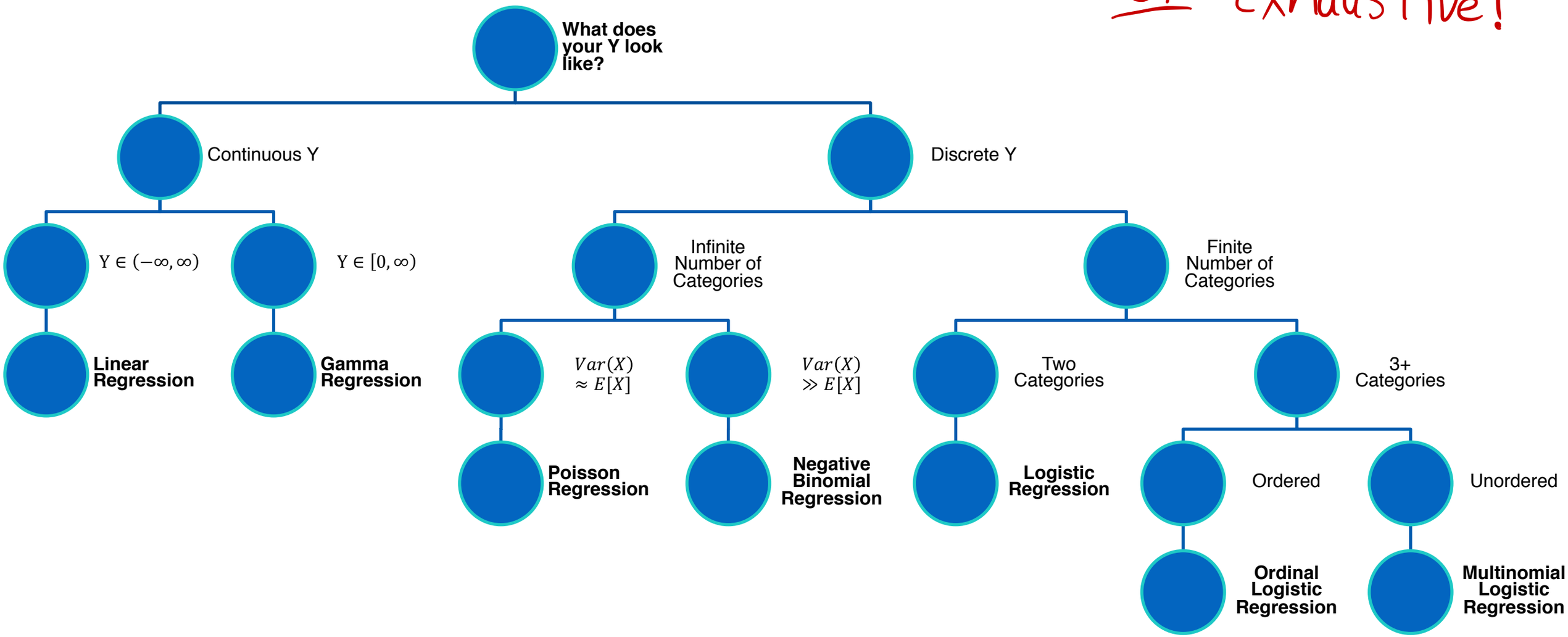
time

↳ how long until ___ happens?

Gamma = $\sum$ Exponential

GENERALIZED LINEAR MODELS CHEAT SHEET

- not exhaustive!



ITERATIVELY REWEIGHTED LEAST SQUARES

- When we fit a linear regression model using ordinary least-squares, the values of $\hat{\boldsymbol{\beta}}$ are directly calculated.

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

- However, there exists no closed-form solution for $\hat{\boldsymbol{\beta}}$ with a generalized linear model.
  - Instead, we must calculate a first guess $\hat{\boldsymbol{\beta}}_1$, update our weights, calculate $\hat{\boldsymbol{\beta}}_2$, and iterate until $|\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{i+1}|$ is sufficiently small.

CONVERGED

$${}_1\hat{\beta} = (X^T W_1 X)^{-1} X^T W_1 y \Longrightarrow {}_2\hat{\beta} = (X^T W_2 X)^{-1} (X^T W_2 y)$$

WHY USE GLMS?

- Generalized linear models are a series of models that:
  - Allow us to easily infer relationships between X and Y.
  - Have convenient parametric distributions.
  - Are flexible enough to handle a variety of scenarios.
    - Hierarchical models.
    - "Zero-inflated" or "zero-truncated" models.



too many zeros



no zeros

30% sales on line

70% sales in person